Build a Human Classifier

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Build a Human Classifier

> Prepare Build a Human Classifier: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Construct a **classifier** with a stated **input** format, selected rule and recorded **output**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-1/14, ai-2/10. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Learning from Examples

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate

Reference example, not the only acceptable wording: Threshold 3 predicts short, tall, tall, tall, disagreeing only at 4 cm: count 1. Threshold 7 predicts short, short, short, tall, disagreeing only at 6 cm: count 1.

Reset the tally for each candidate.

Repeating Efficiently

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a loop by testing its condition before every repetition

Reference example, not the only acceptable wording: Tests: 2 true, 4 true, 6 true, 8 false. Three additions are three iterations. The loop reports 8; the fourth test does not run the body.

Features: Useful Clues

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

Look for: Choose recorded properties that preserve distinctions needed for a stated task

Reference example, not the only acceptable wording: Use length and width measured before the decision. State the order as length in cm, then width in cm; the record is 12, 8.

The target is fits or too large under the guide, and this card is fits because both equalities are included.

Colour and card number are not needed for this rule; the final inspection sticker copies the answer and is unavailable before inspection.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

Concept to explain

Classification maps features to a category using a **decision boundary**.

Term	Meaning
classifier	process that predicts a category
feature space	map whose positions represent feature values
decision boundary	place where predicted class changes

Term	Meaning
borderline case	input near or across competing regions

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Students construct a paper procedure around a fitted threshold, joining **input** checks, conditions, stopping and recorded **output**. Separate **input validation**, predictive correctness and subsequent human action.

Tower1's range and whole-number requirement are classroom restrictions, not general requirements for **classifier**s.

Readiness answer

At 5 predict short; at 6 predict tall. Check that a height was supplied. A blank is missing information, not the number zero.

Input cards with 1, 5, 6, 9, 10 and 11 cm, a blank, 60 mm and 4.5 cm

Blank paper for numbered steps or branch diagrams and a record with **input**, version, outcome and reason

Separate Ribbon1 task cards; no accounts, devices or personal measurements required

End goal for the pictorial case

By the end, I can build a complete **classifier** that validates **input**s before predicting.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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Explain the mechanism

> Model Build a Human Classifier using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT creature features

2. Process — the deciding step

PROCESS compare the feature point with a boundary

3. Output — the result

OUTPUT category A, category B, or not sure

Key idea

CORE IDEA Classification maps features to a category using a **decision boundary**.

How the mechanism works

UNDER THE HOOD A **classifier** partitions **feature space** into decision regions. A **decision boundary** is where the predicted category changes.

Different boundaries may fit the same **training** examples but behave differently on unseen or borderline **input**s.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE Mystery creatures can be sorted using visible features such as antenna count and spot pattern, with the decision rule drawn on a feature map.

Step 1

STEP 1 — OBSERVE creature features

Step 2

STEP 2 — TRACE compare the feature point with a boundary

Step 3

STEP 3 — CONCLUDE category A, category B, or not sure

Model answer

MODEL REASONING Plot each creature by the selected features and draw a line or rule separating labeled examples. Cases near the line are sensitive to small changes and should receive careful review or a not-sure **output**.

A claim to avoid

DO NOT OVERCLAIM A drawn boundary describes the **classifier**'s decision, not a natural law about the objects.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

A human **classifier** uses has wheels means vehicle on a bicycle, wheelchair, toy car and boat.

The teaching case

Apply the rule and identify a limitation.

Exact explanation to model

Why the result follows

The first three receive vehicle; the boat does not. If the intended category includes boats, the rule misses them. Define the category independently of the **prediction** rule.

Explain the added worked example

Explain the added worked example · 1

Your task is to build a paper **classifier** that another learner can operate without guessing your instructions. It will predict short or tall for imaginary towers.

Use the threshold of 6 centimetres selected after adding the height 5 short example in Week 13. Name this selected model Tower1.

Explain the added worked example · 2

Write an **input contract**: one height recorded as a whole number of centimetres from 1 through 10, including both endpoints. Require the unit cm to be shown. This is a classroom design choice about what the procedure accepts.

It is not a claim that every accepted height appeared in **fitting** or that towers outside this range cannot exist.

A complete **classifier** needs instructions for checking the **input**, applying Tower1 when the **input** is accepted, and recording the outcome. It must also say what happens when the **input** fails a check.

The threshold alone does not provide these surrounding instructions.

For every record, first check that a height is present and its stated unit is cm. Then check that it is a whole number from 1 through 10.

If any check fails, record **no prediction** and the reason, then stop **process**ing that record. Do not continue into the threshold comparison.

Explain the added worked example · 3

If all checks pass, compare the height with 6. At least 6 gives the **prediction** tall; otherwise give short. Record the original **input**, **model version** Tower1 and the **prediction**.

For a rejected record, record the original **input**, Tower1, **no prediction** and a reason. Tower1 identifies the intended model even when its threshold was not used.

Key terms

Term	Meaning
Classifier	A procedure that assigns a predicted category to an input
Input contract	The format and conditions an input must satisfy for a procedure to accept it
Validation	Checking whether an input satisfies the stated contract
No prediction	An explicit outcome when the classifier does not assign a category
Model version	A name that identifies the particular model used for a prediction

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Input	Contract result	Outcome
1 cm	Accepted at lower endpoint	Predict short
9 cm	Accepted	Predict tall
10 cm	Accepted at upper endpoint	Predict tall
0 cm	Outside 1 through 10	No prediction
60 mm	Unit is not cm	No prediction
4.5 cm	Not a whole number	No prediction

A valid 9 cm **input** receives a tall **prediction**, but the **input** check does not establish that the **prediction** matches its comparison label. **Validation** checks the contract.

Evaluation checks **prediction**s against appropriate **evidence**. Keep these questions separate.

Model the reasoning aloud · 4

Sixty millimetres equals six centimetres, but this version accepts cm records only. It does not contain a conversion step.

Ask for a correctly recorded cm measurement or deliberately design and check a conversion step in a new version. Do not silently treat the digits 60 as centimetres. A missing height must not become an invented zero either.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Build a Human Classifier

A fictional ribbon sorter uses a fixed instruction card called RibbonR.

The facts we will use

Supplied facts

Valid **input**: whole-number length from 0 to 15 with unit cm.

Supplied facts

Predict long at least 9 cm; otherwise short. Missing, wrong-unit or out-of-range values request clarification.

AI ACADEMY / AI-1/14

Build a Human Classifier

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Input contract

Check that length is present, whole, within 0-15, and in cm.



2. Valid cases

8 cm -> short; 9 cm -> long.



3. Invalid cases

16 cm and blank length -> clarification, not a forced label.



4. Keep records

Store input, contract result, rule version and prediction.

CHANGE ONE THING

Provide length 9 with unit mm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Input contract	Check that length is present, whole, within 0–15, and in cm.
2. Valid cases	8 cm → short; 9 cm → long.
3. Invalid cases	16 cm and blank length → clarification, not a forced label.
4. Keep records	Store input, contract result, rule version and prediction.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **prediction** boundary answers a different question from **input validation**. Length 9 passes the contract and meets the long boundary.

Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

What this example does not establish

Keep the stated limit

Contract validity does not prove the predicted category agrees with a checked reference label.

Change one thing in the pictured case

Change one thing

Change: Provide length 9 with unit mm.

Predicted result

Prediction: Request clarification rather than predicting long.

Why the result follows

Reason: RibbonR requires cm; the unit mismatch fails its declared **input contract**.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Build a Human Classifier, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A human **classifier** uses has wheels means vehicle on a bicycle, wheelchair, toy car and boat.

2. State the decision to explain

Write the question to explain: Apply the rule and identify a limitation.

3. Reveal the worked reasoning

Reveal after **prediction**s: The first three receive vehicle; the boat does not. If the intended category includes boats, the rule misses them. Define the category independently of the **prediction** rule.

Hold back the independent task

Keep for an independent attempt: Use a human **classifier**: three straight sides forming a closed shape means triangle; four equal straight sides and four right angles forming a closed shape means square. Classify a three-sided card and a card whose corner is hidden. When should you pause?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Check presence, cm unit, whole number and range 1 through 10 before comparing. At 1, 6 and 10 cm predict short, tall and tall. At 11 cm record **no prediction**, outside range; at blank record **no prediction**, missing height.

Record Tower1 throughout. Acceptance of 10 cm does not verify its label. Exit: 9 cm passes the contract and predicts tall, but correctness still needs **evidence**. A blank must stop as **no prediction**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve threshold equality and the difference between missing and zero. Introduce the goal of instructions another learner can execute.
7 to 20 minutes	Read the Tower1 contract and model aloud. Trace 5 cm, 6 cm and blank through the diagram. Stop physically before comparing the rejected card.

Stage	Teacher action
20 to 33 minutes	Compare valid boundaries and rejected units or fractions. Complete W1 and explain why acceptance is not proof of a correct label.
33 to 45 minutes	Start W2 in pairs. One learner writes, the other follows only the written instructions. Require an explicit stop after a failed check.
Session 2 start 0 to 8 minutes	Retrieve the output record and both paths. Ask whether no prediction is a tower size.
8 to 23 minutes	Use the guided task and W3 to test partners' procedures at endpoints, threshold and missing input. Record and repair ambiguity.
23 to 38 minutes	Collect W4 independently before support. Look for transfer to the new range, threshold, labels and version instead of copied Tower1 constants.
38 to 45 minutes	Use W5 and the exit to separate contract checks from correctness evidence. Preview final tests kept apart from model selection.

Reduce help deliberately

Completion practice

Completion task: Write numbered instructions for Tower1, including **input** checks, the stopping path after a failed check, the threshold rule and the record to keep.

Your partner must be able to use them without supplying an unstated rule.

Reasoning to listen for

Teacher check: Check present height and cm unit, then whole number in 1 through 10. On failure, record original **input**, Tower1, **no prediction** and the reason; stop that record.

Otherwise predict tall if height is at least 6, short otherwise. Record original **input**, Tower1 and **prediction**. Equivalent clear branches are acceptable.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Constructing a **classifier**. Proposed explanation: “Any numeric **input** must receive a category”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: Check the stated unit, whole-number condition and supported range before comparing the threshold.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Trace 0 cm and explain why clarification is different from predicting short.

Reasoned teacher answer

0 is a valid whole-number centimetre **input**, so RibbonR predicts short. Clarification means no supported **prediction** was made because required **input** checks failed.

Changed-case reasoning: Request clarification rather than predicting long. RibbonR requires cm; the unit mismatch fails its declared **input contract**.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the Build a Human Classifier investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Classify mystery creatures and make the boundary visible.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR A **classifier** must always choose one of its labels.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Any numeric **input** must receive a category: Check the stated unit, whole-number condition and supported range before comparing the threshold.

An accepted **input** proves a correct **prediction**: Ask separately whether the **input** satisfies the contract and whether the **prediction** agrees with evaluation **evidence**.

No prediction means a third category of tower: It is a system outcome indicating that no short or tall category was assigned.

Writing a rejection is enough even if **process**ing continues: Follow the failed path to an explicit stop so the later rule cannot overwrite the outcome.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Build a Human Classifier questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

The first three receive vehicle; the boat does not. If the intended category includes boats, the rule misses them. Define the category independently of the **prediction** rule.

Independent question

Independent question · withhold the key

Use a human **classifier**: three straight sides forming a closed shape means triangle; four equal straight sides and four right angles forming a closed shape means square. Classify a three-sided card and a card whose corner is hidden. When should you pause?

Expected independent answer

Teacher key · Expected independent answer

The visible closed three-straight-sided card is a triangle. A square requires four equal sides and four right angles; equal sides alone are insufficient. Pause on the partly hidden card when the required sides or angles cannot be checked. Ask for an unobscured view instead of guessing.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A **classifier** must always choose one of its labels.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> #### Teacher key · Original transfer answer
>
> A bat and toy aeroplane are not birds but may satisfy the wing rule; a penguin is a bird even though it cannot fly. Use reliable biological labels and gather more relevant features.
>
> Accept explicit uncertainty where the available **evidence** is insufficient.

Explanations for the additional workbook examples

W1 Write an input contract

Workbook question · W1

Write the complete Tower1 **input contract**. Give one accepted example and three different rejected examples, each with its reason. Explain why an accepted **input** does not have to be one of the examples used to fit the threshold.

Teacher reasoning · W1

Require a present height, unit cm, a whole number and range 1 through 10 inclusive. For example, 7 cm is accepted; blank is missing, 60 mm has the wrong unit and 11 cm is outside the range.

Fractional values such as 4.5 cm also fail. The contract defines accepted **input**s; it does not list only the **fitting** records or establish **prediction** correctness.

W2 Construct both outcome paths

Workbook question · W2

Write numbered instructions for Tower1, including **input** checks, the stopping path after a failed check, the threshold rule and the record to keep. Your partner must be able to use them without supplying an unstated rule.

Teacher reasoning · W2

Check present height and cm unit, then whole number in 1 through 10. On failure, record original **input**, Tower1, **no prediction** and the reason; stop that record. Otherwise predict tall if height is at least 6, short otherwise.

Record original **input**, Tower1 and **prediction**. Equivalent clear branches are acceptable.

W3 Trace boundaries and a broken path

Workbook question · W3

Trace 1 cm, 5 cm, 6 cm, 10 cm and 11 cm using Tower1. A faulty procedure records **no prediction** for a blank, then continues and treats the blank as zero to report short.

Identify the missing instruction and the proper final outcome.

Teacher reasoning · W3

The first four are accepted and predict short, short, tall and tall. At 11 cm reject for being outside the range. The faulty path lacks stopping after failure; a blank must end as **no prediction** with reason missing height.

It must not be converted to zero or reach the threshold rule.

W4 Build an independent ribbon classifier

Workbook question · W4

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise.

Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for **no prediction** and the **model version** in each record.

Teacher reasoning · W4

Check present length, cm unit, whole number and range 1 through 12; otherwise record **no prediction** with reason and stop. For accepted **input**s, use at least 8 for long, otherwise short.

Ribbon1 at 7 cm predicts short; at 8 cm predicts long; blank gives **no prediction** because length is missing; 13 cm gives **no prediction** because outside range.

Each record includes original **input**, Ribbon1 and outcome, plus reason when rejected.

W5 Explain what your procedure establishes

Workbook question · W5

Ribbon1 accepts 12 cm and predicts long. No comparison label has been supplied for that record. State what has been checked and what remains unknown.

Explain why changing a blank to 1 cm just to produce a category would be a mistake.

Teacher reasoning · W5

The value satisfies the **input contract** and the stated rule predicts long. Agreement with an appropriate comparison label has not been checked.

Replacing a blank with 1 invents a measurement, conceals missing information and violates the stated rejection path. Ask for the measurement instead.

Assess independent transfer

Independent question · withhold the key

Selected case: Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise.

Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for **no prediction** and the **model version** in each record.

Selected case answer

> **Teacher key · Selected case answer**
>
> Check present length, cm unit, whole number and range 1 through 12; otherwise record **no prediction** with reason and stop. For accepted **input**s, use at least 8 for long, otherwise short.
>
> Ribbon1 at 7 cm predicts short; at 8 cm predicts long; blank gives **no prediction** because length is missing; 13 cm gives **no prediction** because outside range.
>
> Each record includes original **input**, Ribbon1 and outcome, plus reason when rejected.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Build a Human Classifier to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Make the sorter's current **decision boundary** visible as a rule chart or feature map.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 15: Save New Examples for Testing. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use separate check, compare and record cards with arrows. Let partners move each **input** physically and read the branch aloud.

Extension: propose a separate mm-to-cm conversion stage; state what extra cases would need checking, including a conversion that yields a fractional centimetre.

W4 is secure with both paths, range 1 through 12, threshold 8 including equality, explicit stops and Ribbon1 records. After support retry Flag1: whole cm from 1 through 9, unit cm, large at least 5 otherwise small.

At 4, 5, blank and 10 cm expect small, large, **no prediction** for missing and **no prediction** for range.

Week 15 keeps final evaluation examples unseen during **fitting** and explains why repeated changes based on their labels weaken the independence of that test.

Plan the delayed return

Suggested return points in this level: ai-1/15, ai-1/17, ai-1/20. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Build a Human Classifier lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Construct a **classifier** with a stated **input** format, selected rule and recorded **output**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Write numbered instructions for Tower1, including **input** checks, the stopping path after a failed check, the threshold rule and the record to keep.

Your partner must be able to use them without supplying an unstated rule.

Supported practice answer

Reasoned practice answer: Check present height and cm unit, then whole number in 1 through 10. On failure, record original **input**, Tower1, **no prediction** and the reason; stop that record.

Otherwise predict tall if height is at least 6, short otherwise. Record original **input**, Tower1 and **prediction**. Equivalent clear branches are acceptable.

Check every learner before discussion: Use the worked case for Constructing a **classifier**. Proposed explanation: “Any numeric **input** must receive a category”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Check the stated unit, whole-number condition and supported range before comparing the threshold.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Provide length 9 with unit mm. Require a **prediction** and a reason before comparing. Expected result: Request clarification rather than predicting long.

Reason: RibbonR requires cm; the unit mismatch fails its declared **input contract**.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

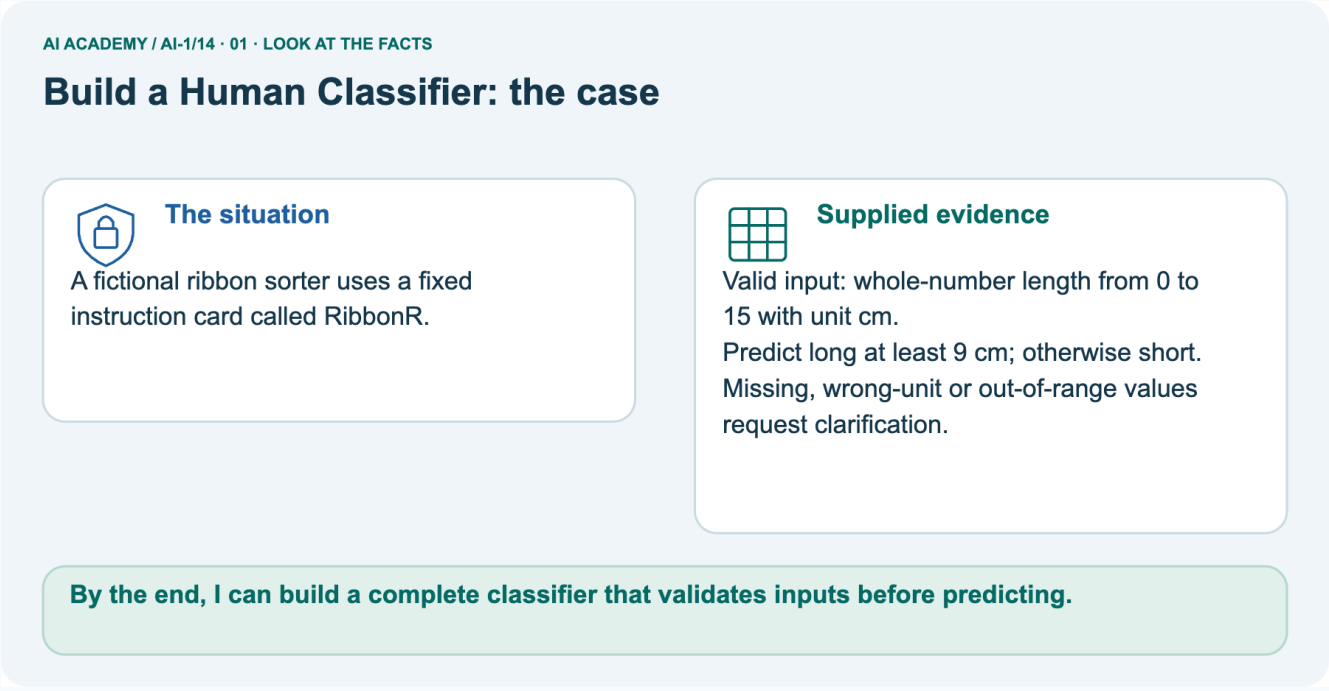
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Visual concept journey

> Show one Build a Human Classifier graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Build a Human Classifier: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional ribbon sorter uses a fixed instruction card called RibbonR. Supplied facts: Valid **input**: whole-number length from 0 to 15 with unit cm. Predict long at least 9 cm; otherwise short.

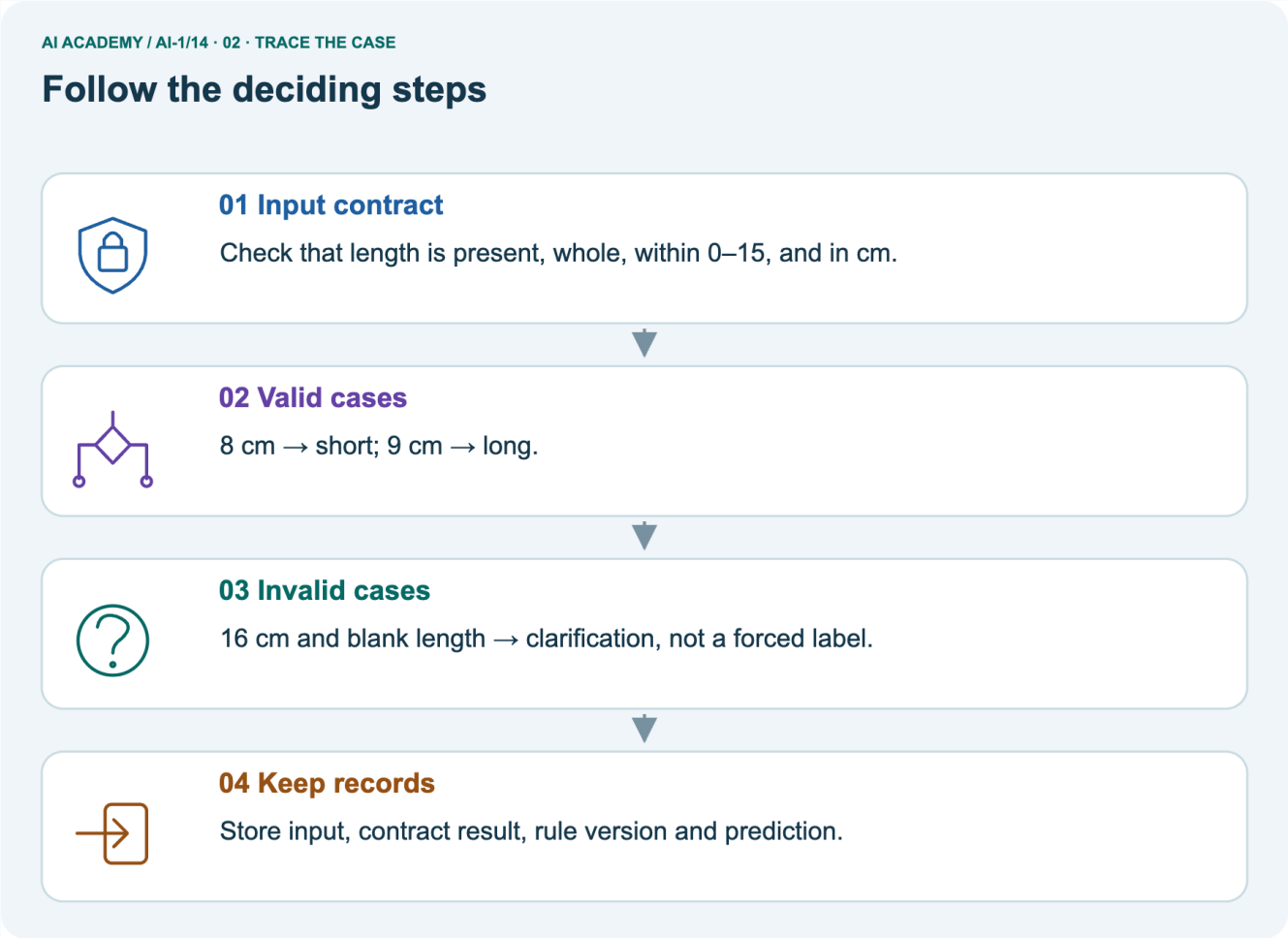
Missing, wrong-unit or out-of-range values request clarification.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

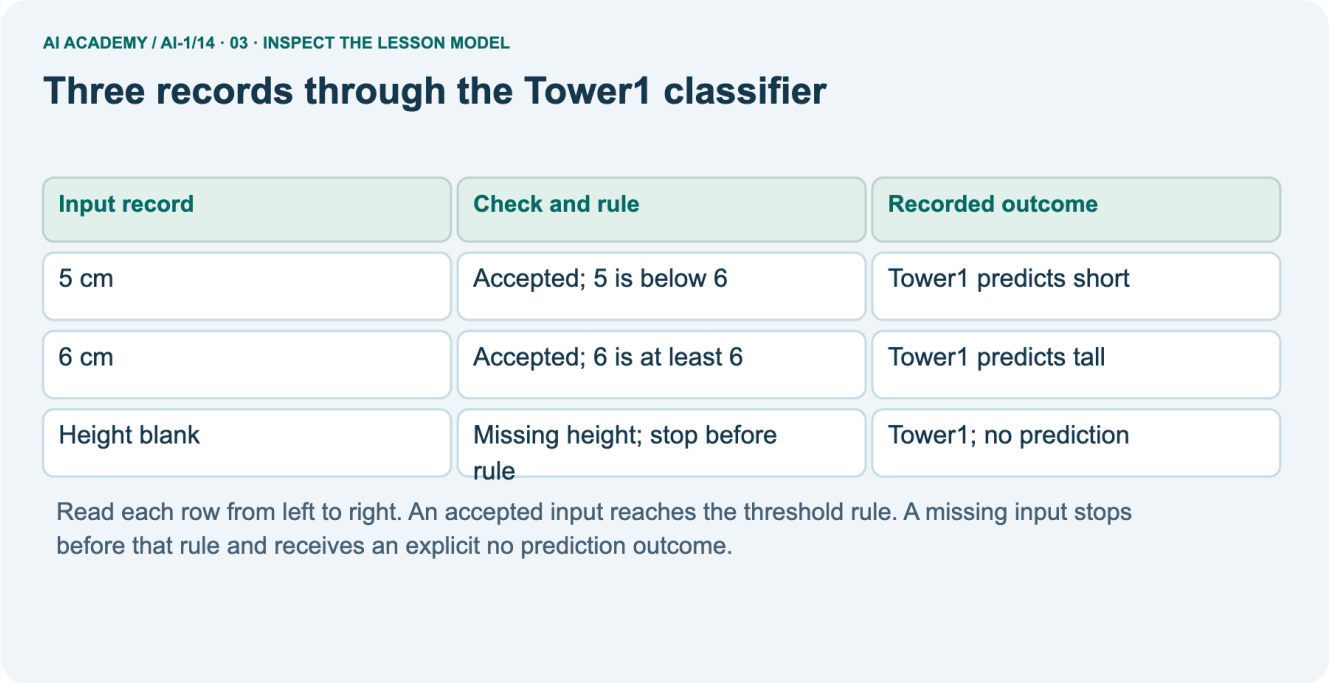
Input contract: Check that length is present, whole, within 0–15, and in cm. → Valid cases: 8 cm → short; 9 cm → long. → Invalid cases: 16 cm and blank length → clarification, not a forced label. → Keep records: Store **input**, contract result, rule version and **prediction**.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Three records through the Tower1 classifier



Follow the labelled picture with the matching explanation. Open the vector diagram

Three records through the Tower1 **classifier**. 5 cm | Accepted; 5 is below 6 | Tower1 predicts short. 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall.

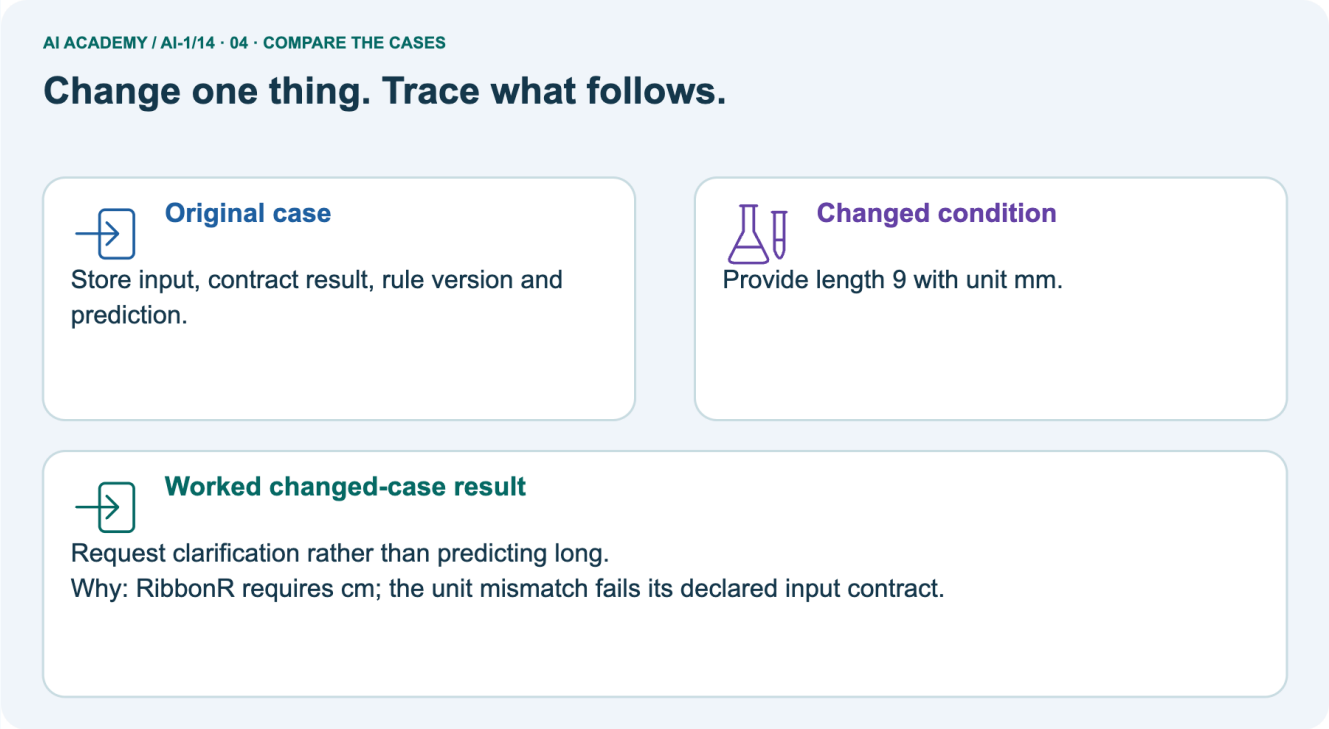
Height blank | Missing height; stop before rule | Tower1; **no prediction**

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Store **input**, contract result, rule version and **prediction**. Changed condition: Provide length 9 with unit mm. Expected result: Request clarification rather than predicting long.

Reason: RibbonR requires cm; the unit mismatch fails its declared **input contract**.

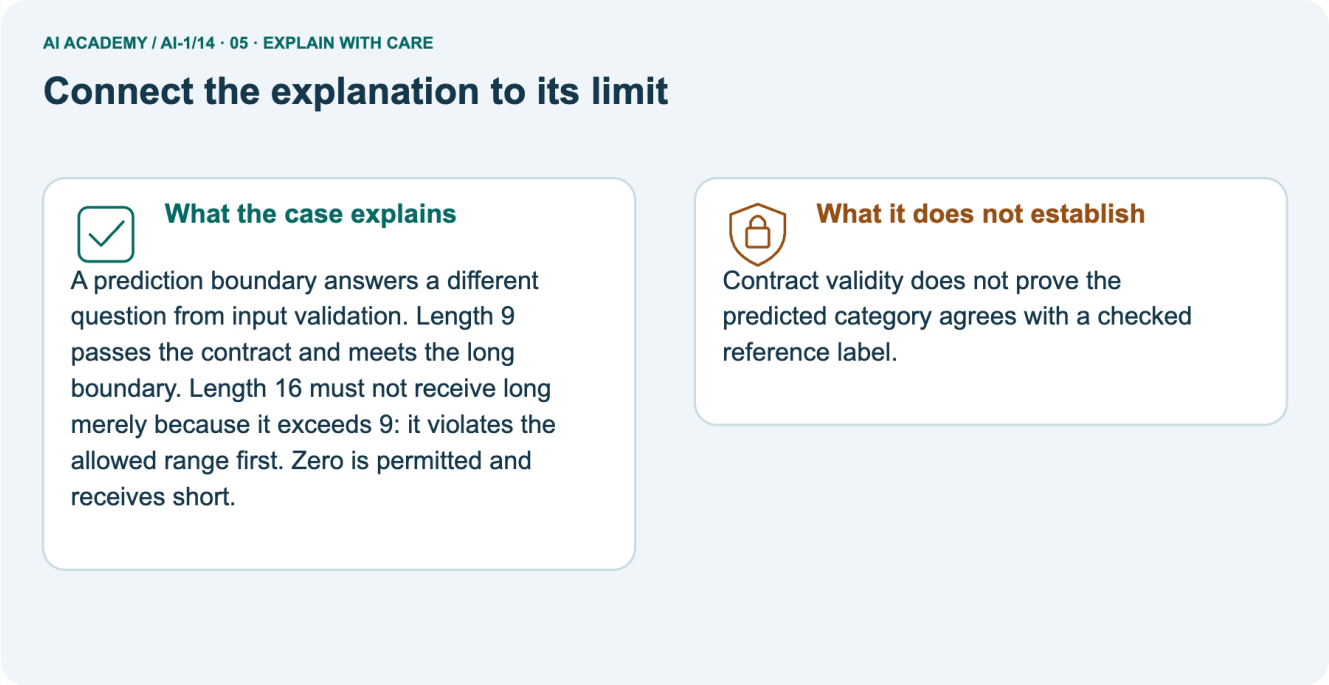
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: RibbonR requires cm; the unit mismatch fails its declared **input contract**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **prediction** boundary answers a different question from **input validation**. Length 9 passes the contract and meets the long boundary.

Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

Limit: Contract validity does not prove the predicted category agrees with a checked reference label.

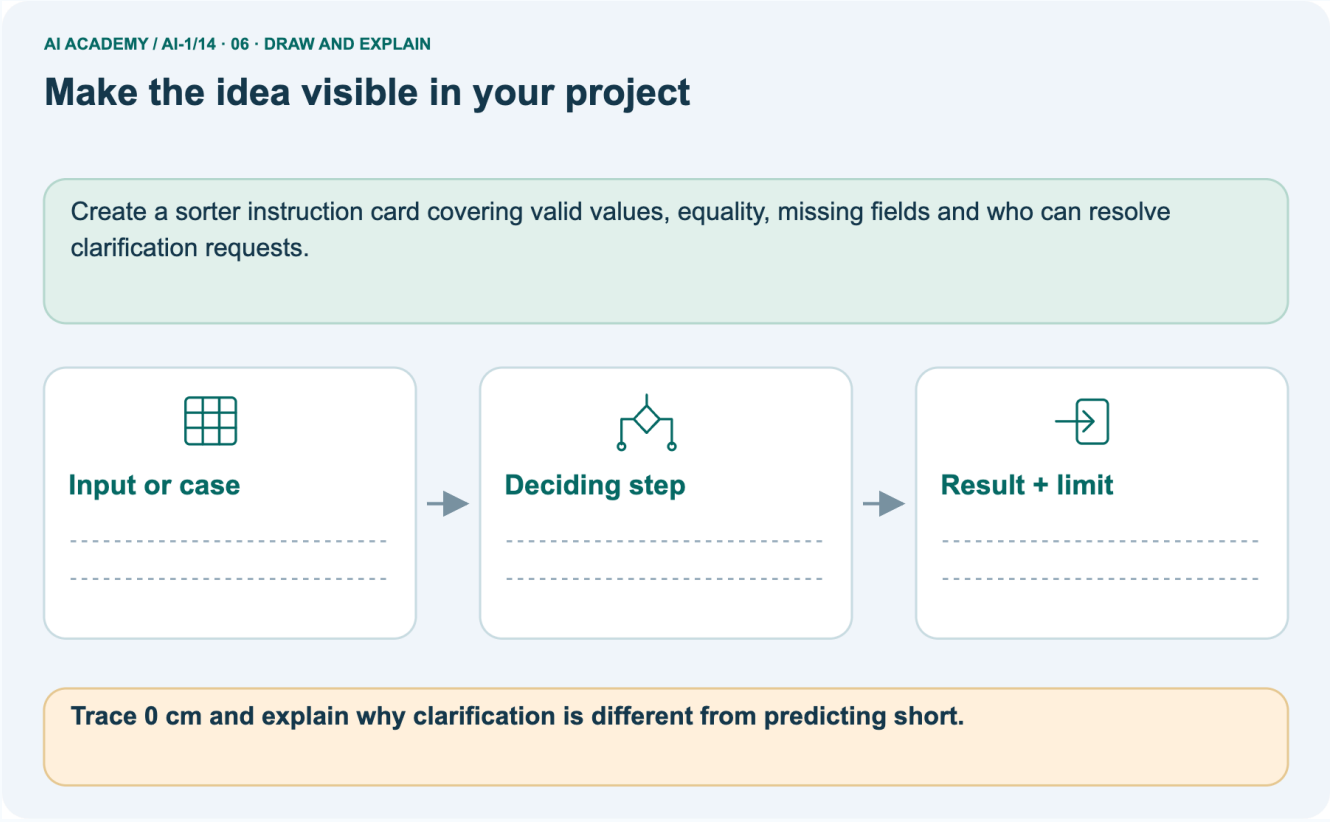
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Contract validity does not prove the predicted category agrees with a checked reference label.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests. Draw your **input**, deciding step, result and limit.

Supported question: Trace 0 cm and explain why clarification is different from predicting short.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 0 is a valid whole-number centimetre **input**, so RibbonR predicts short. Clarification means no supported **prediction** was made because required **input** checks failed.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.